

Q.14

Linear Regression :-

$$(x_1, y_1) = (1, 40) (3, 65) (5, 75)$$

①

calc. $\bar{x}$ mean $= \frac{1+3+5}{3}$

$$\boxed{\bar{x} = 3}$$

$$= \frac{9}{3} = 3$$

y mean $= \frac{40+65+75}{3}$

$$\boxed{\bar{y} = 60}$$

② calc. slope

$$w = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Numerator:- $[(1-3)(40-60) + (3-3)(65-60) + (5-3)(75-60)]$
 $= [(-2)(-20) + (0) + (2)(15)]$
 $= 40 + 30 = 70$

Denominator $= \sum (x_i - \bar{x})^2$
 $= (1-3)^2 + (3-3)^2 + (5-3)^2$
 $= 2^2 + 0 + 2^2$
 $= 4 + 4 = 8$

$$w = \frac{70}{8} = 8.75$$

③ $\boxed{b = \bar{y} - w \cdot \bar{x}}$

$$b = 60 - (8.75) \times 3$$

$$= 60 - 26.25 = 33.75$$

③

Q.15

(3)

$$\hat{y} = wx_i + b$$

$$\boxed{\hat{y} = 8.75x + 33.75}$$

Q.15

Using the linear regression

$$\hat{y} = wx_i + b$$

$$w = 8.75$$

$$b = 33.75$$

So, For - Student who has studied zero hours.

$$\hat{y} = (8.75) \times 0 + 33.75$$

$$= 0 + 33.75$$

$$\boxed{\hat{y} = 33.75}$$

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Q.12

Probability of losing 5 toss
in a row.

Probability of losing one toss = out of 1
 $= \frac{1}{2}$

2 toss in row $= \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right)$

5 toss in row $= \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right)$
 $= \left(\frac{1}{2}\right)^5$

$= 0.031$

$P(5 \text{ losses in row}) = 0.031$

Q.13

Rolling of Six face dice twice

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

All possible sums:-

2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

= 11

All possible sums which we can get by rolling die twice is = 11

Expectation

$$E(X) = \frac{1 \times 1 + 2 \times 2 + 3 \times 3 + 4 \times 4 + 5 \times 5 + 6 \times 6}{6}$$

$$E(X) = 7$$

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Q.16

⑦ Gradient descent.

$$f(x) = x^4 - 12x^2$$

Gradient descent update rule

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t)$$

$$f(x) = x^4 - 12x^2$$

$$\begin{aligned} \nabla f(x) &= 4x^3 - (2)(12)x \\ &= 4x^3 - 24x \end{aligned}$$

$$\nabla^2 f(x) = (3)(4)x^2 - 24$$

$$f'(x) = 4x^3 - 24x$$

⑧ $x_0 = 0, \eta = 0.1$

$$\nabla^2 f(x) = 12x^2 - 24$$

$$x_1 = x_0 - \eta \nabla f(x_0)$$

$$= 0 - (0.1)(0 - 24)$$

$$= 0 - (0.1)(-24)$$

$$= 0 - (-0.4)$$

$$= 0 + 0.4$$

$$= 0.4$$

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⑤

⑥

④

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(b) $x_0 = 0.1 \quad \eta = 0.1$

$$\begin{aligned} &= x_k - \eta (12x^2 - 4) \\ &= (0.1) - (0.1) (12 \times (0.1)^2 - 4) \\ &= (0.1) - [(0.1) (-3.88)] \\ &= [0.1 - (-0.388)] \\ &= -\cancel{0.288} \quad 0.488 \end{aligned}$$

(c) $x_0 = -0.1 \quad \eta = 0.1$

$$\begin{aligned} &= x_k - \eta (12x^2 - 4) \\ &= (-0.1) - (0.1) (12 \times (0.1)^2 - 4) \\ &= (-0.1) - (0.1) [-3.88] \\ &= -0.1 + 0.388 \\ &= 0.288 \end{aligned}$$

(d) for gradient descent, local minimum becomes global minimum
 $\| \nabla_{\theta} L(\theta) \| \approx 0$ (gradient near zero)

$\nabla L(\theta)$ points in direction of steepest increase
 - negative $\nabla L(\theta)$ means steepest decrease, we move in direction to reduce the loss.

①

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Q.4

*

Vectors linear independence:-
= $\{(1,2) (1,3) (1,4)\}$

Vectors are set to be linearly independent

when

$\sum \beta_i a_i = 0$ for β not all zero
that mean $[a_1 v_1 + a_2 v_2 + a_3 v_3 = 0]$

$$C = \beta_1 a_1 + \beta_2 a_2 + \beta_3 a_3$$

if vector can be represented as ~~scat~~ linear
multiplication / combination of other two vectors
then they are called as independent
vectors.

E.g.

$(1,2)$

$(1,3)$

$(1,4)$

Let's scale 1st 2 vectors by $k=1, k=2$

$k=1$

$(1,2)$

$(1,3)$

$(1,4)$

$k=2$

$(2,4)$

$(2,6)$

$(2,8)$

$k=3$

$(3,6)$

$(3,9)$

$(3,12)$

so ~~no~~ vector can be represented as combination
of other 2 vectors.

Hence

these vectors are

linearly independent

Relation:-

$$(1,2) - 2(1,3) + (1,4) = (0,0)$$

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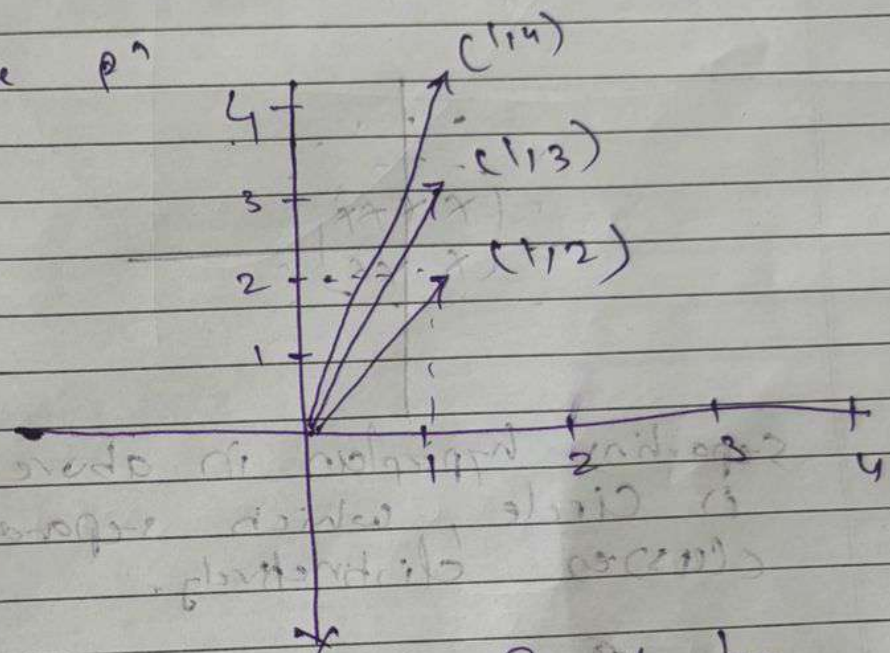
(2)

Q.5

Subspace

Vectors $(1, 2)$
 $(1, 3)$
 $(1, 4)$

subspace P^1



that 3 vectors can be formed by
transforming other vectors
hence linearly dependent.

$(0, 0)$

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Q.3)

$$f(x_1, x_2) = x_1^2 + x_2^2$$

$$\text{Condition: } x_1 + 2x_2 \leq -1$$

Step 1:-

Form the Lagrangian

$$L(x_1, x_2, \lambda) = x_1^2 + x_2^2 - \lambda(x_1 + 2x_2 + 1)$$

$$\text{Take } \frac{\partial L}{\partial x_1} = 2x_1 - \lambda(x_1)$$

$$2x_1 - \lambda = 0$$

$$2x_1 = \lambda$$

$$\boxed{x_1 = \frac{\lambda}{2}}$$

$$\text{Take } \frac{\partial L}{\partial x_2} = 2x_2 - \lambda(2x_2)$$

$$\boxed{x_2 = \frac{\lambda}{2} = -\lambda}$$

$$x_1 + 2x_2 + 1 \leq 0$$

$$\left(\frac{\lambda}{2}\right) + 2\left(\frac{\lambda}{4}\right) + 1 \geq 0$$

$$\frac{\lambda}{2} + \frac{\lambda}{2} + 1 \geq 0$$

$$\frac{2\lambda + 2}{2} \geq -1$$

$$\boxed{\lambda \geq -1}$$

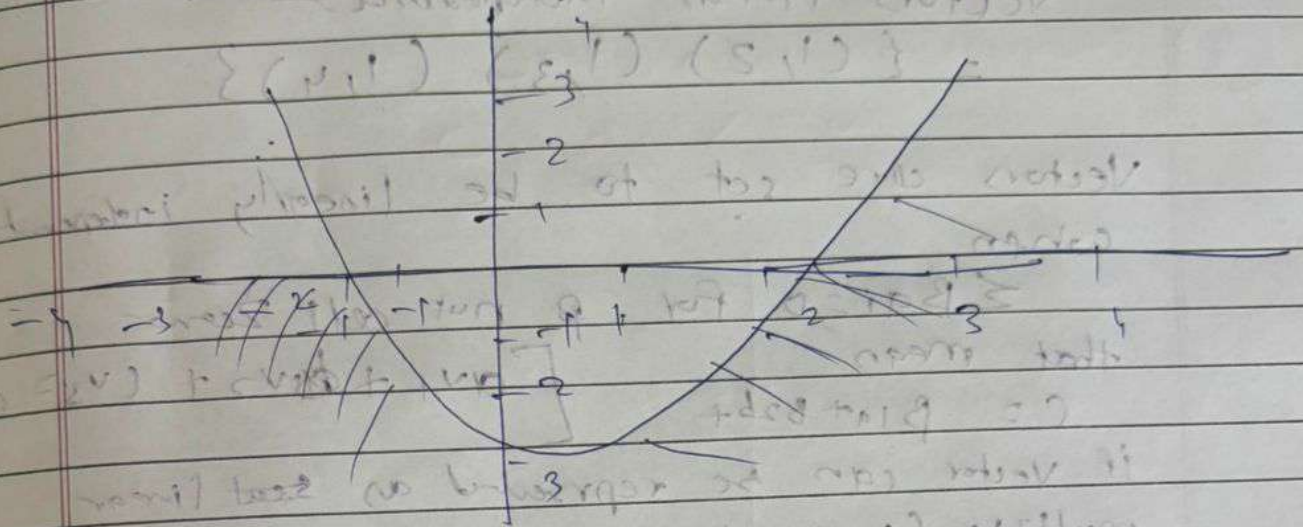
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Region:-



$$(x_1, x_2) = \begin{pmatrix} \frac{2}{5} \\ -\frac{1}{5} \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$2(x_1 + 2x_2 - 1) = 0$$

$$-\frac{2}{5} = -2 \cdot 1 = 1 = \frac{-5 \cdot 2}{5} = 1$$

$$\boxed{2x_1 - 2x_2 = 2}$$

$(2, 0)$	$(0, 1)$	$(0, 1)$
$(0, 0)$	$(0, 0)$	$(0, 0)$
$(0, 0)$	$(0, 0)$	$(0, 0)$
$(0, 0)$	$(0, 0)$	$(0, 0)$

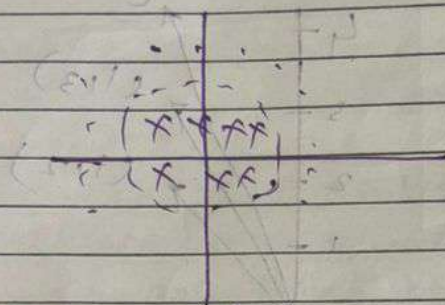
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S.V.M.

Q.6 Inputs = 2 dimensional

(9) Separating hyperplane
 $\langle w, x \rangle + b = 0$



Separating hyperplane in above case
i) Circle, which separates two
classes distinctively.

Equation of circle = $a^2 + b^2 = c^2$

given any two points we can say
if it lies within the circle or not,

Hence, data is linearly separable.

Q.8

$$y_i = \text{sig}(\langle w x_i \rangle + b)$$

$$w = \sum_{i=1}^n \alpha_i x_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j (x_i, x_j)$$

Lagrange α_i

if $\alpha_i = 0$ and $\alpha_i = 0$

(a)

Sample

1	0	0
2	0	0
3	0.5	0
4	0.5	0
5	1	1.5
6	0	0

$\Rightarrow$ Sample - 1, 2, 3, 4 are correctly classified.